

Translate shapes

Use the scaffold, show each step and ask for an adult check when marked.

Name: _____ Date: _____

1. Read the model: Translate $(4, -2)$ by 3 left and 5 up. Copy or complete the first step.

2. Reflect $(-6, 2)$ in the y-axis.

3. A point becomes $(7, -4)$ after moving 9 right and 3 down. Find its start.

4. Miro says: Miro changes x and y in the same way without checking the direction or mirror line. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: Translate $(4, -2)$ by 3 left and 5 up. Copy or complete the first step.
Answer $(1, 3)$.
2. Reflect $(-6, 2)$ in the y-axis.
Answer $(6, 2)$.
3. A point becomes $(7, -4)$ after moving 9 right and 3 down. Find its start.
Answer $(-2, -1)$.
4. Miro says: Miro changes x and y in the same way without checking the direction or mirror line. Is this correct? Explain.
Answer Track horizontal and vertical changes separately and apply the exact transformation to every point.

Translate shapes

Work independently. Show a complete method and check at least one answer.

Name: _____ Date: _____

1. Translate (4, -2) by 3 left and 5 up.

6. Check this result using an inverse, estimate or known fact: A point becomes (7, -4) after moving 9 right and 3 down. Find its start.

2. Reflect (-6, 2) in the y-axis.

7. Prove or explain your reasoning: Plot A(-4, 3). Which quadrant contains A?

3. A point becomes (7, -4) after moving 9 right and 3 down. Find its start.

8. Identify and correct this misconception: Miro changes x and y in the same way without checking the direction or mirror line.

4. Solve this independently and show every stage: Translate (4, -2) by 3 left and 5 up.

9. Write a complete sentence explaining how you checked one answer.

5. Use a second method or representation for: Reflect (-6, 2) in the y-axis.

10. Write and solve a similar question to: A point becomes (7, -4) after moving 9 right and 3 down. Find its start.

Teacher answers and checking prompts

1. Translate (4, -2) by 3 left and 5 up.
Answer (1, 3).
2. Reflect (-6, 2) in the y-axis.
Answer (6, 2).
3. A point becomes (7, -4) after moving 9 right and 3 down. Find its start.
Answer (-2, -1).
4. Solve this independently and show every stage: Translate (4, -2) by 3 left and 5 up.
Answer (1, 3).
5. Use a second method or representation for: Reflect (-6, 2) in the y-axis.
Answer Expected result: (6, 2). Method or representation may vary.
6. Check this result using an inverse, estimate or known fact: A point becomes (7, -4) after moving 9 right and 3 down. Find its start.
Answer Expected result: (-2, -1). A valid check must be shown.
7. Prove or explain your reasoning: Plot A(-4, 3). Which quadrant contains A?
Answer Quadrant II. The explanation must show why the method works.
8. Identify and correct this misconception: Miro changes x and y in the same way without checking the direction or mirror line.
Answer Track horizontal and vertical changes separately and apply the exact transformation to every point.
9. Write a complete sentence explaining how you checked one answer.
Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.
10. Write and solve a similar question to: A point becomes (7, -4) after moving 9 right and 3 down. Find its start.
Answer Answers vary. The new question must use the same mathematical structure and include a correct solution.

Translate shapes

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: _____ Date: _____

1. Prove or explain your reasoning: Plot A(-4, 3). Which quadrant contains A?

6. Create a new example that tests the same idea as: A point becomes (7, -4) after moving 9 right and 3 down. Find its start.

2. Prove the answer to this question, rather than only calculating it: Translate (4, -2) by 3 left and 5 up.

7. Find a second method or representation. Compare its efficiency with your first method.

3. Solve this in two different ways and compare them: Reflect (-6, 2) in the y-axis.

8. Write one always, sometimes or never statement about today's learning and justify it.

4. Work backwards from the answer to reconstruct the question: (-2, -1).

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.

5. Explain why this reasoning fails: Miro changes x and y in the same way without checking the direction or mirror line.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.

Teacher answers and checking prompts

1. Prove or explain your reasoning: Plot A(-4, 3). Which quadrant contains A?

Answer Quadrant II. The explanation must show why the method works.
2. Prove the answer to this question, rather than only calculating it: Translate (4, -2) by 3 left and 5 up.

Answer Expected result: (1, 3). Proof or justification must match the lesson structure.
3. Solve this in two different ways and compare them: Reflect (-6, 2) in the y-axis.

Answer Expected result: (6, 2). Two valid methods and a comparison are required.
4. Work backwards from the answer to reconstruct the question: (-2, -1).

Answer One valid reconstruction is: A point becomes (7, -4) after moving 9 right and 3 down. Find its start. Other equivalent questions are acceptable.
5. Explain why this reasoning fails: Miro changes x and y in the same way without checking the direction or mirror line.

Answer Track horizontal and vertical changes separately and apply the exact transformation to every point.
6. Create a new example that tests the same idea as: A point becomes (7, -4) after moving 9 right and 3 down. Find its start.

Answer Answers vary. The example and solution must preserve the mathematical structure.
7. Find a second method or representation. Compare its efficiency with your first method.

Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.
8. Write one always, sometimes or never statement about today's learning and justify it.

Answer Answers vary. A valid example and counterexample should support the classification.
9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.

Answer Answers vary. The error must be mathematically relevant and the correction must be precise.
10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.

Answer Answers vary. The explanation must distinguish the mathematical structure from the changed value.